RISC-V

Pipelined Architecture

Critical Paths

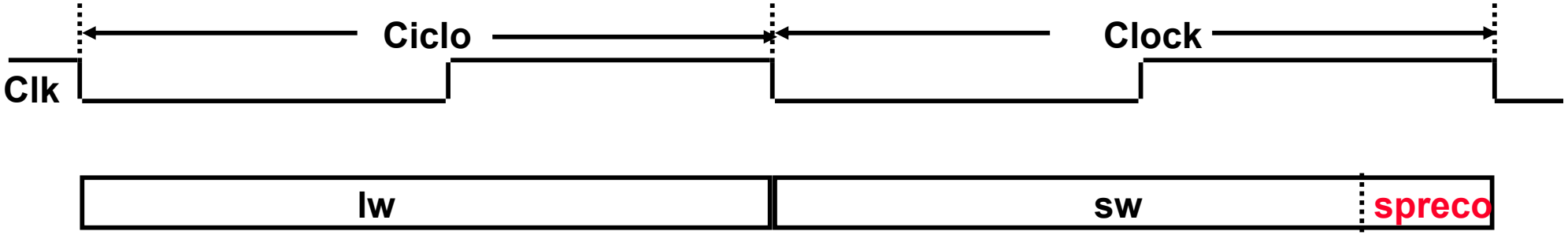
Durata del ciclo di clock nell'ipotesi che sia trascurabile il ritardo in: mux, unità di controllo, estensione in segno, lettura del PC, shift left e fili

- Tempo di accesso nella memoria istruzioni e dati : 200 ps
- Tempo di calcolo in ALU e addizionatori : 200 ps
- Tempo di accesso ai registry : 100 ps

Instr.	I Mem	Reg Rd	ALU Op	D Mem	Reg Wr	Total
R-type	200	100	200		100	600
load	200	100	200	200	100	800
store	200	100	200	200		700
beq	200	100	200			500
jump	200					200

Ciclo di clock singolo: vantaggi e svantaggi

- Uses the clock cycle inefficiently – the clock cycle must be timed to accommodate the slowest instruction
  - especially problematic for more complex instructions like floating point multiply



- May be wasteful of area since some functional units (e.g., adders) must be duplicated since they can not be shared during a clock cycle

but

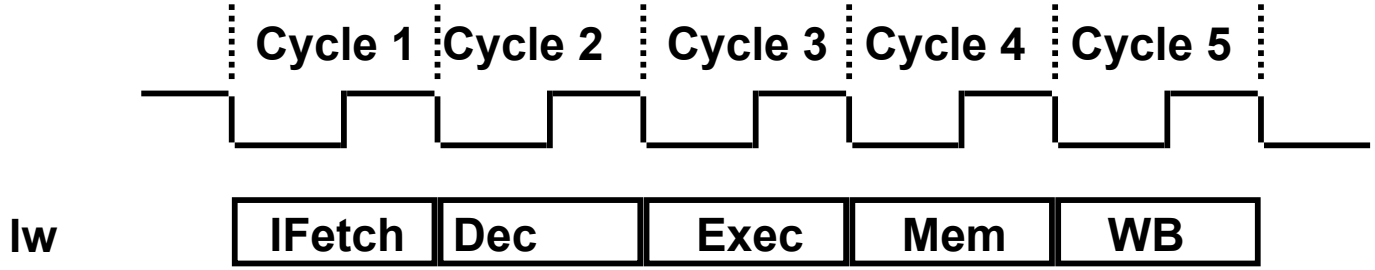
- Is simple and easy to understand

How Can We Make It Faster?

- Start fetching and executing the next instruction before the current one has completed
  - Pipelining – (all?) modern processors are pipelined for performance
  - Remember *the* performance equation:

    CPU time = CPI * CC * IC
- Under *ideal* conditions and with a large number of instructions, the speedup from pipelining is approximately equal to the number of pipe stages
  - A five stage pipeline is nearly five times faster because the CC is nearly five times faster
- Fetch (and execute) more than one instruction at a time
  - Superscalar processing – stay tuned

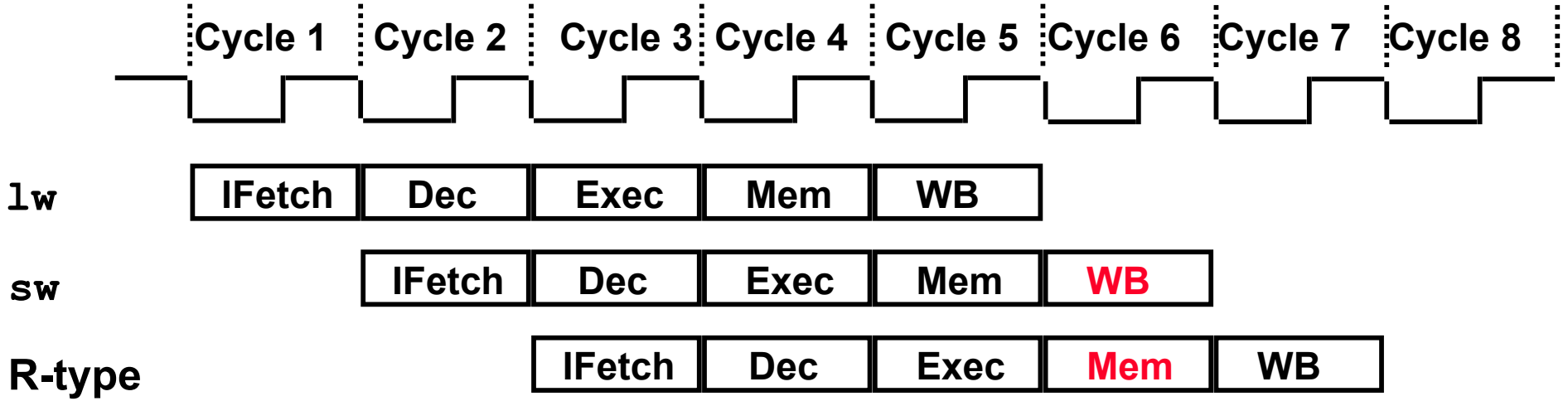
Fasi di esecuzione



- IFetch: lettura istruzione e aggiornamento PC
- Dec: lettura registri e decodifica istruzione
- Exec: esecuzione operazioni artimetiche/logiche
- Mem: lettura/scrittura memoria dati
- WB: scrittura registri

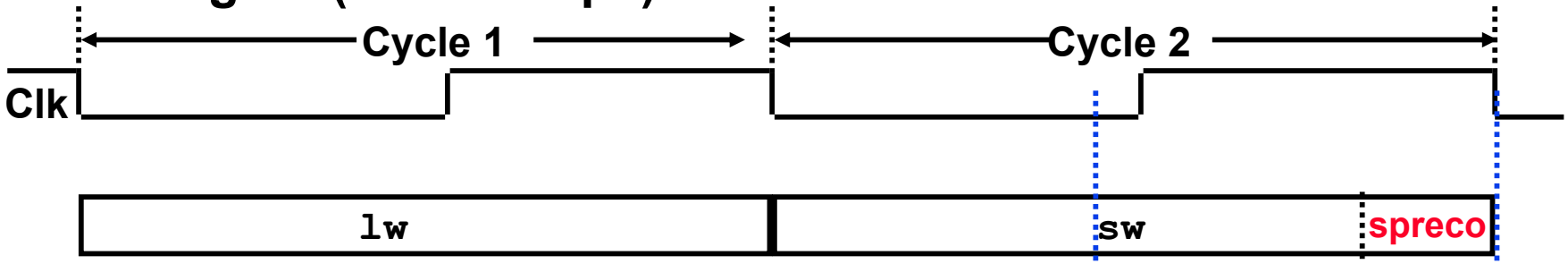
Pipeline

- Iniziare esecuzione prossima istruzione prima del completamento della precedente
  - migliora throughput
  - Non migliora la latenza

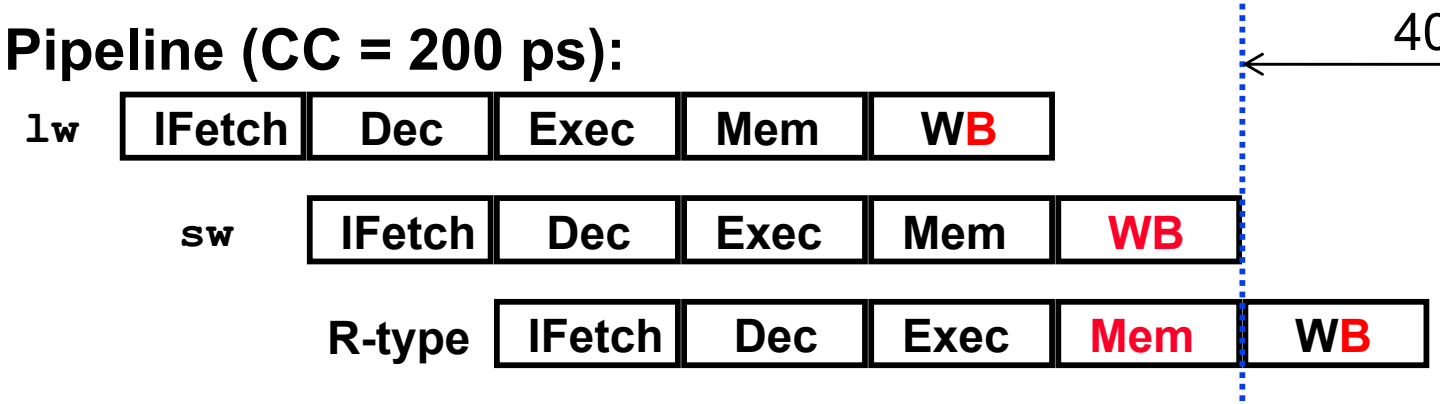


Single Cycle versus Pipeline

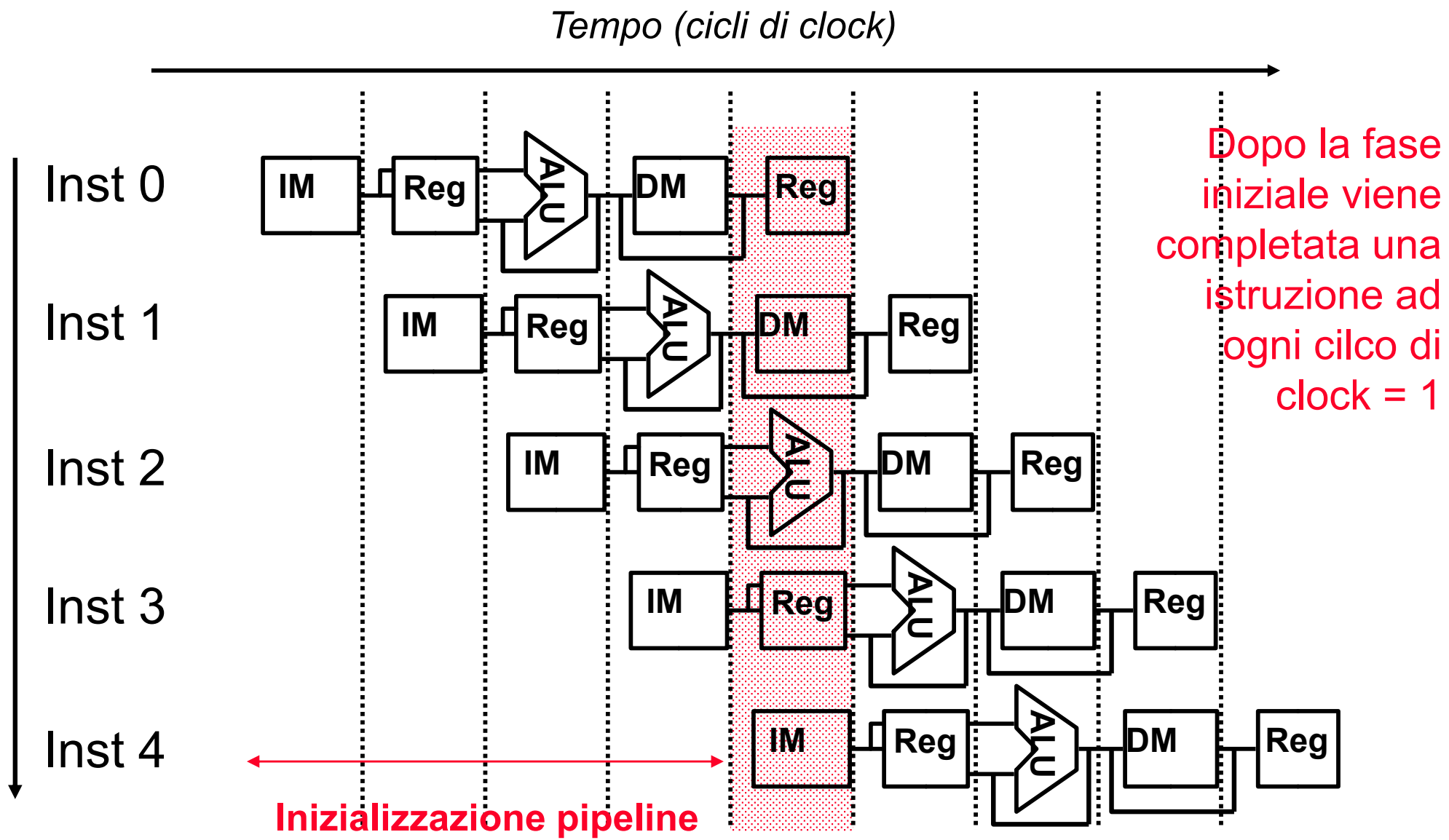
Ciclo singolo (CC = 800 ps):



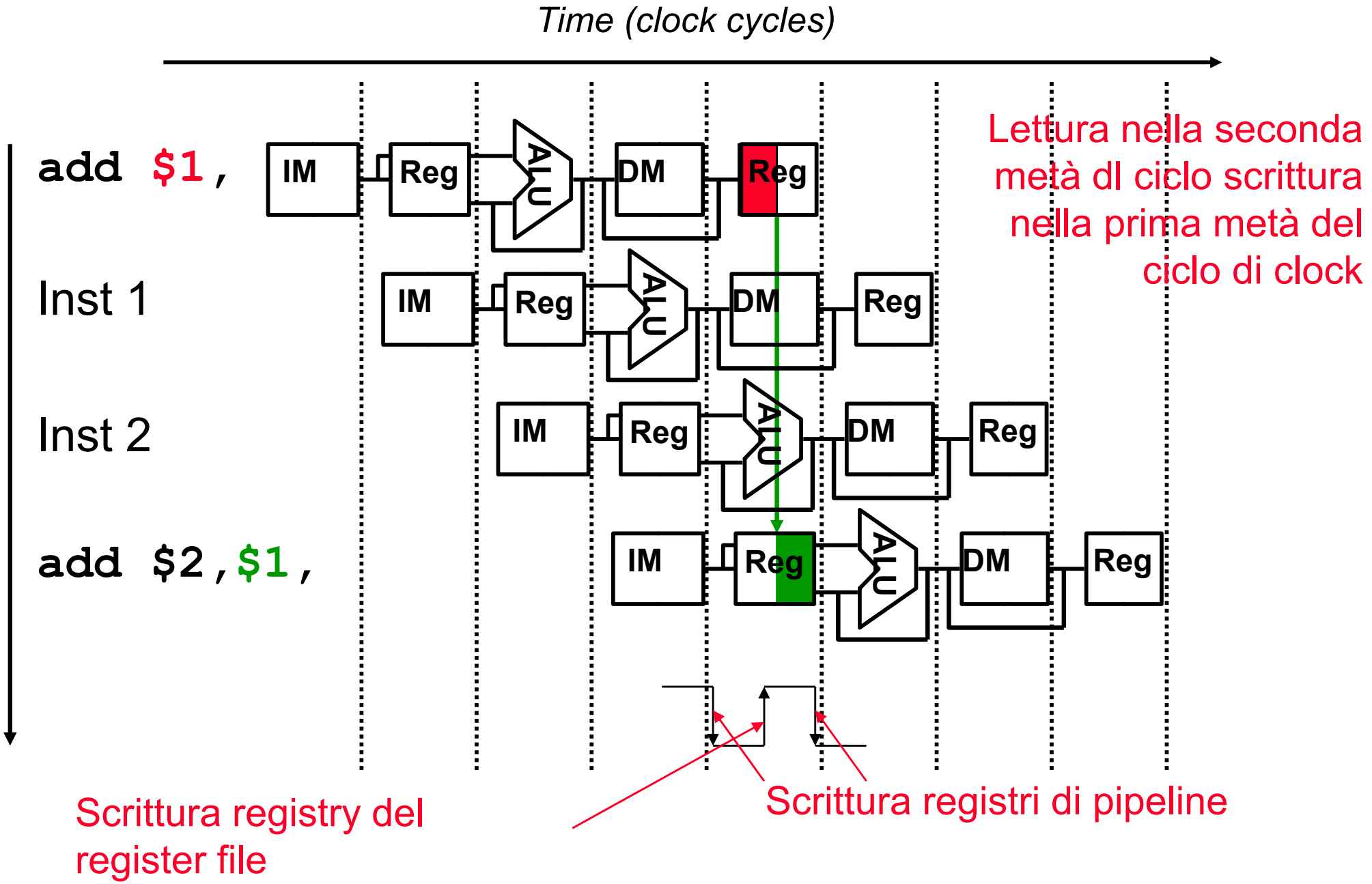
Pipeline (CC = 200 ps):



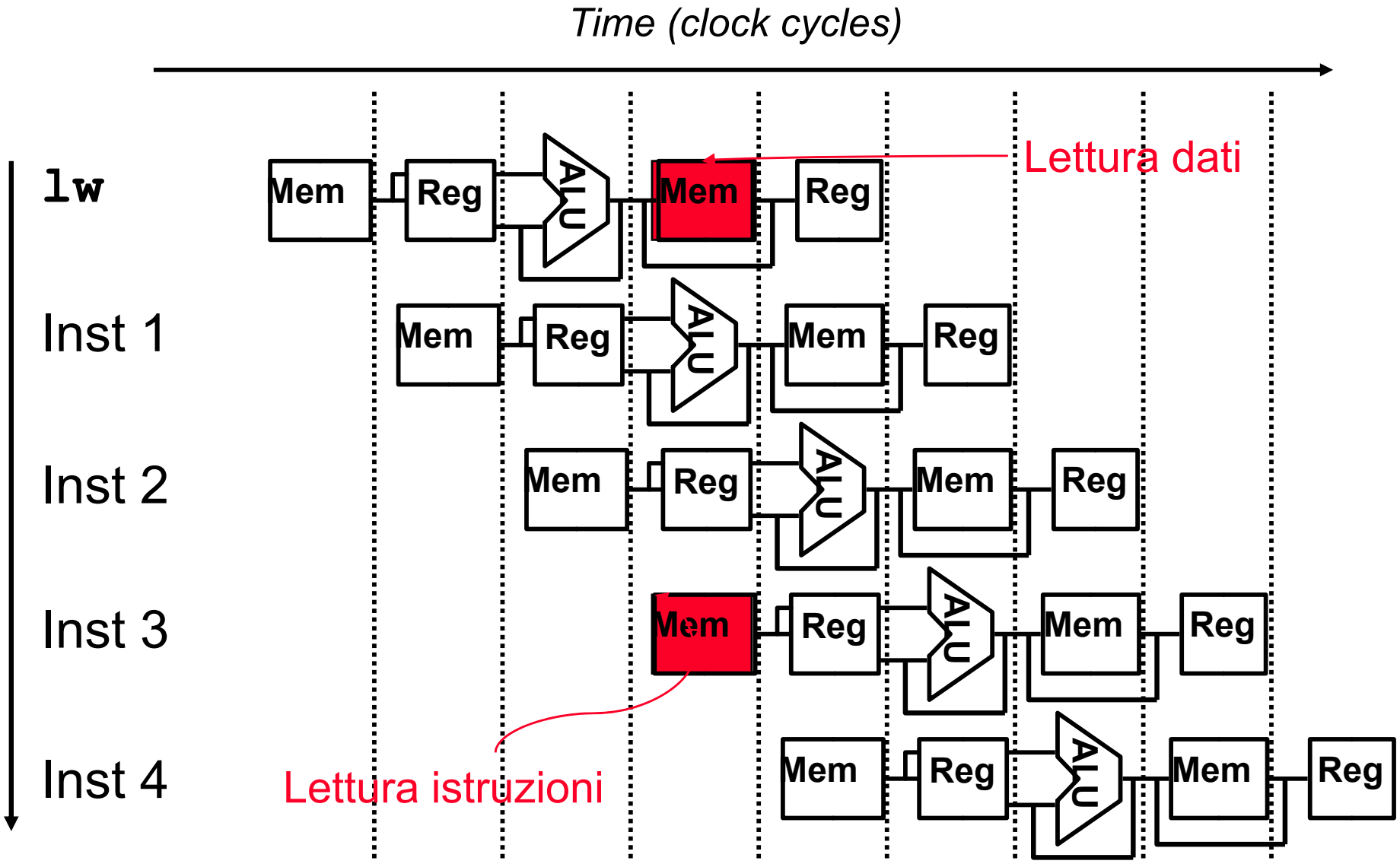
Pipeline


Tempo (cicli di clock)
Inst 0
Inst 1
Inst 2
Inst 3
Inst 4
IM
Reg
ALU
DM
Reg
Dopo la fase iniziale viene completata una istruzione ad ogni cilco di clock = 1
Inizializzazione pipeline

Il Register File provoca conflitti ?



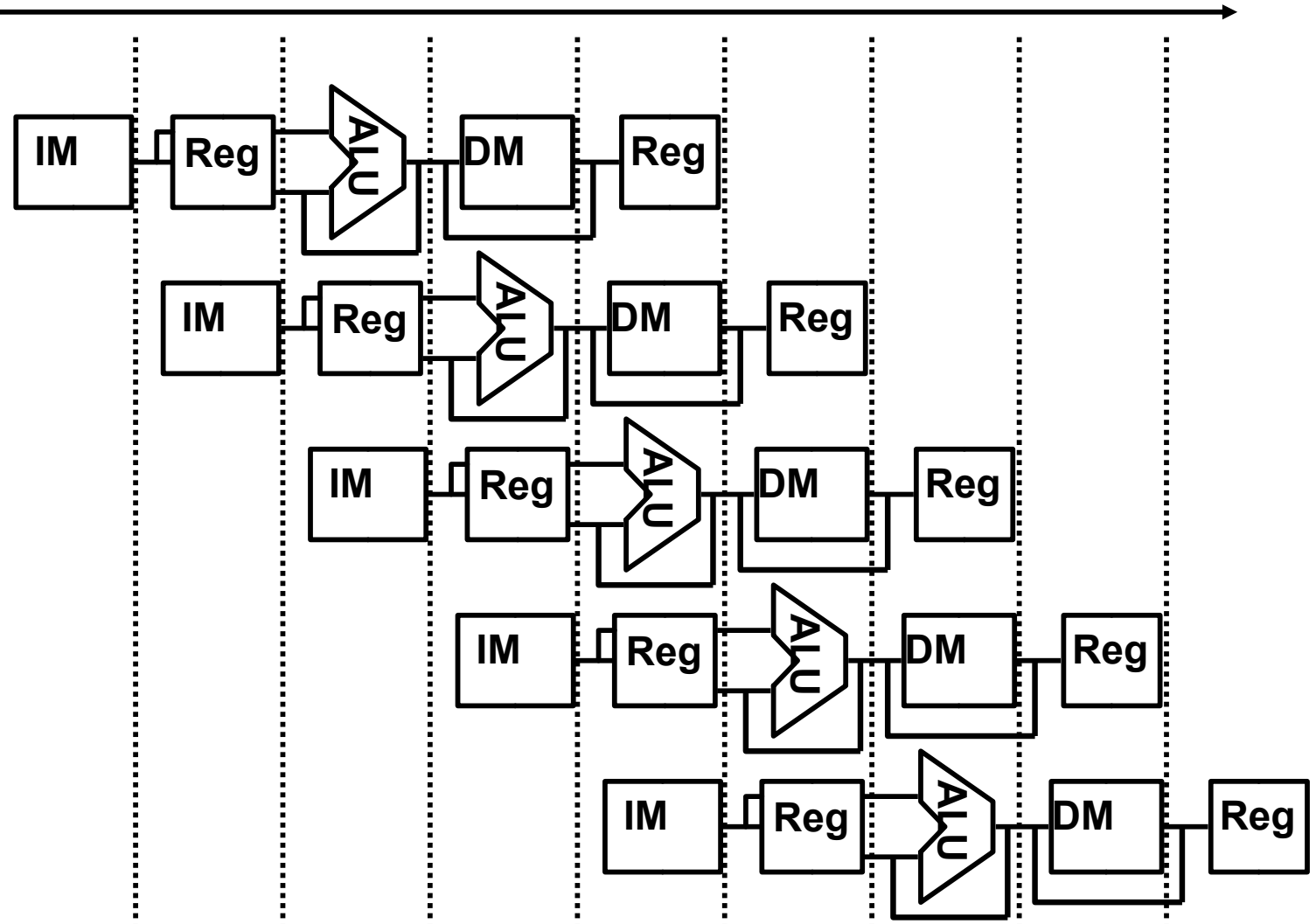
Memoria singola: conflitto strutturale



Data Hazards

Le dipendenze causano conflitti (read before write)

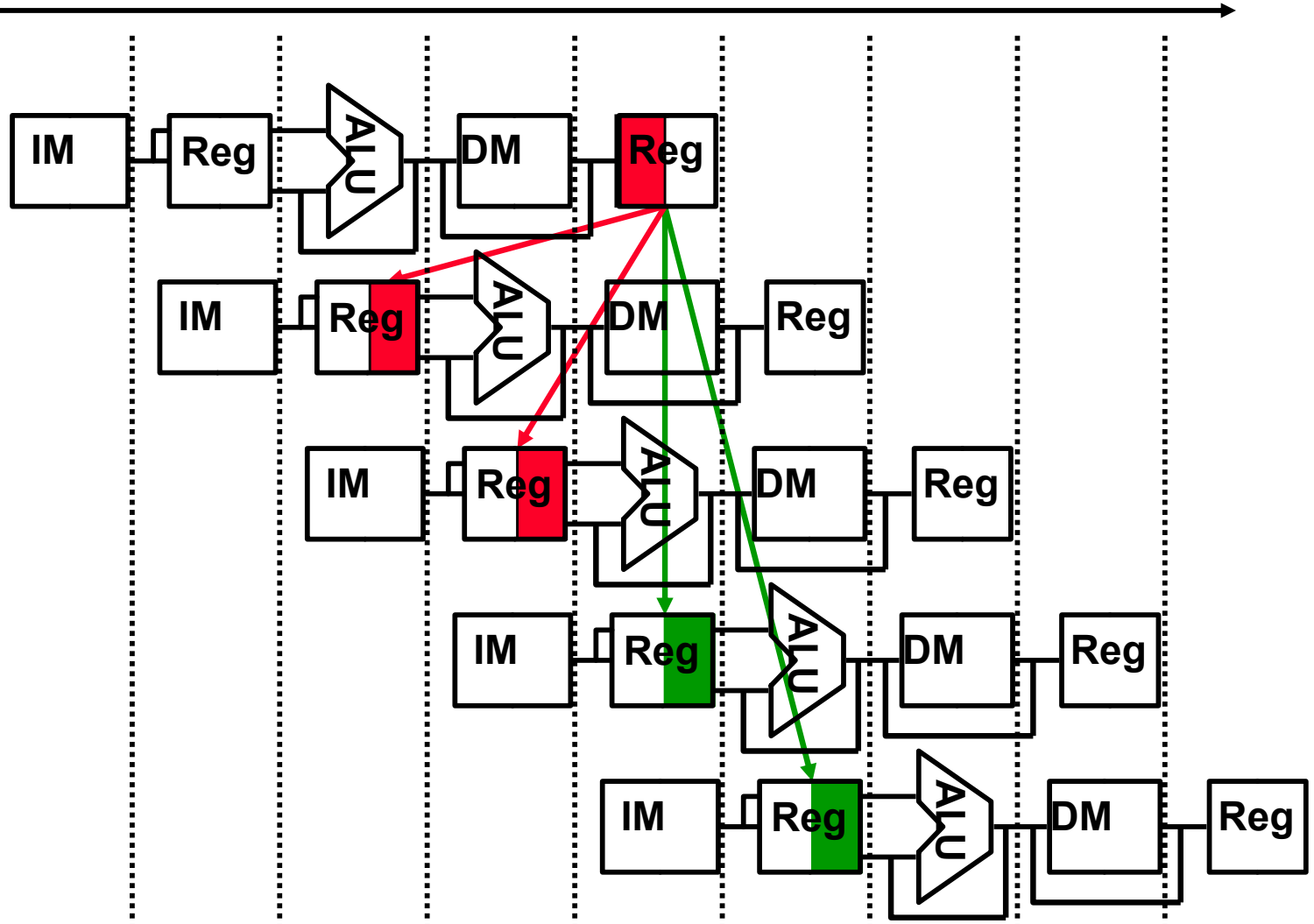
```
add $1,
sub $4,$1,$5
and $6,$1,$7
or  $8,$1,$9
xor $4,$1,$5
```



Data Hazards

Le dipendenze causano conflitti (read before write)

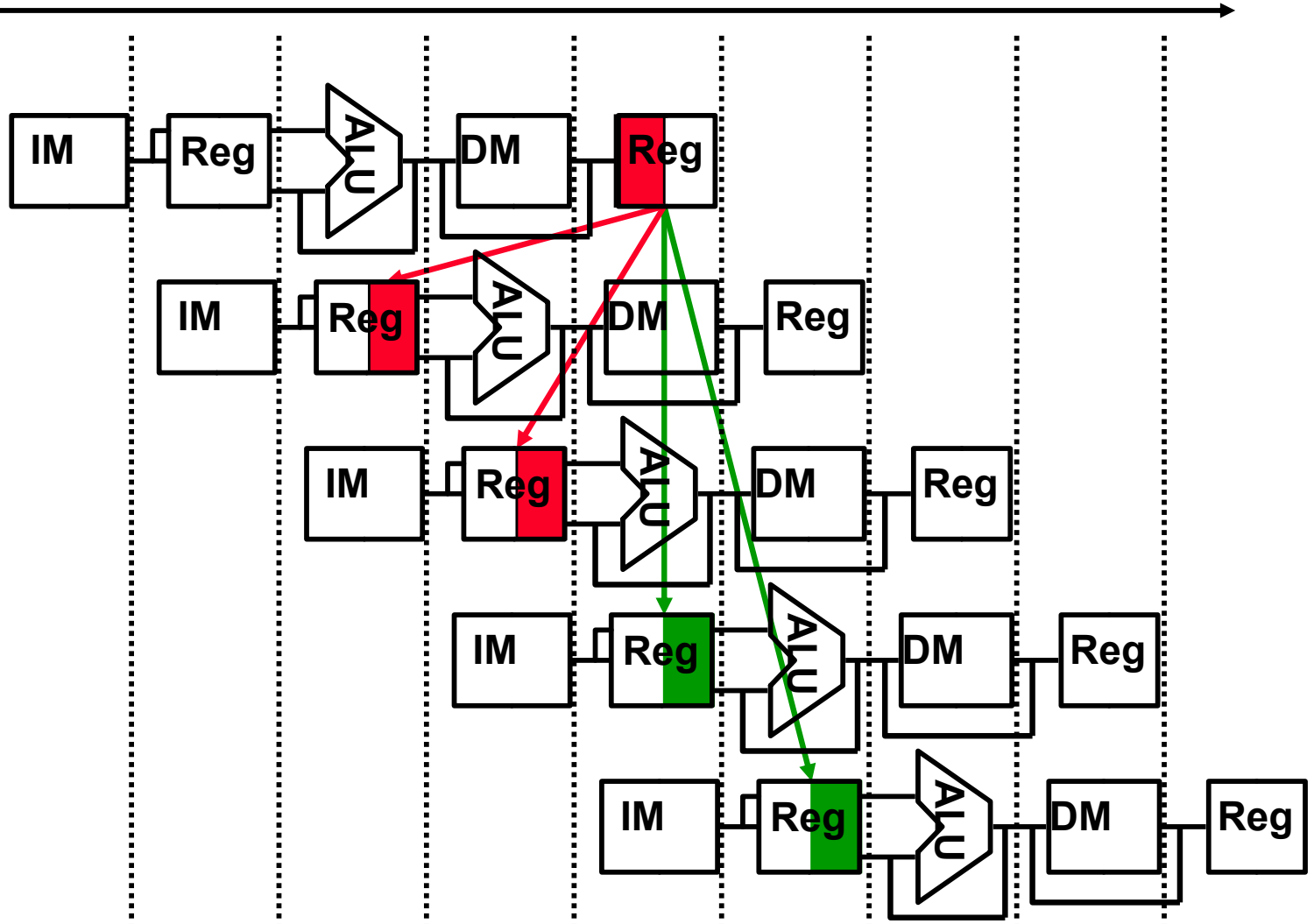
```
add $1,
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and $6,$1,$7
or  $8,$1,$9
xor $4,$1,$5
```



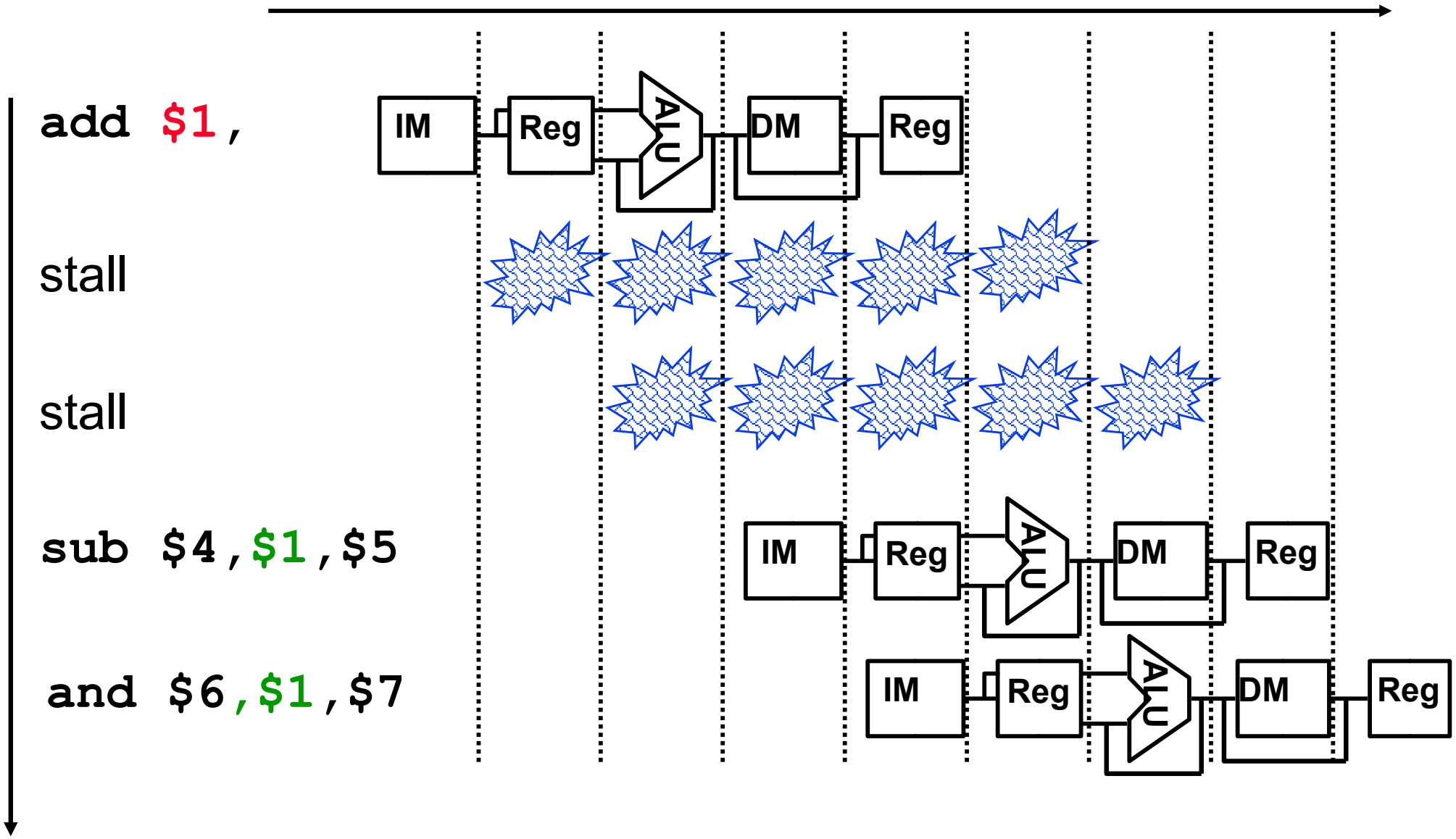
Data Hazards

Le dipendenze causano conflitti (load-use)

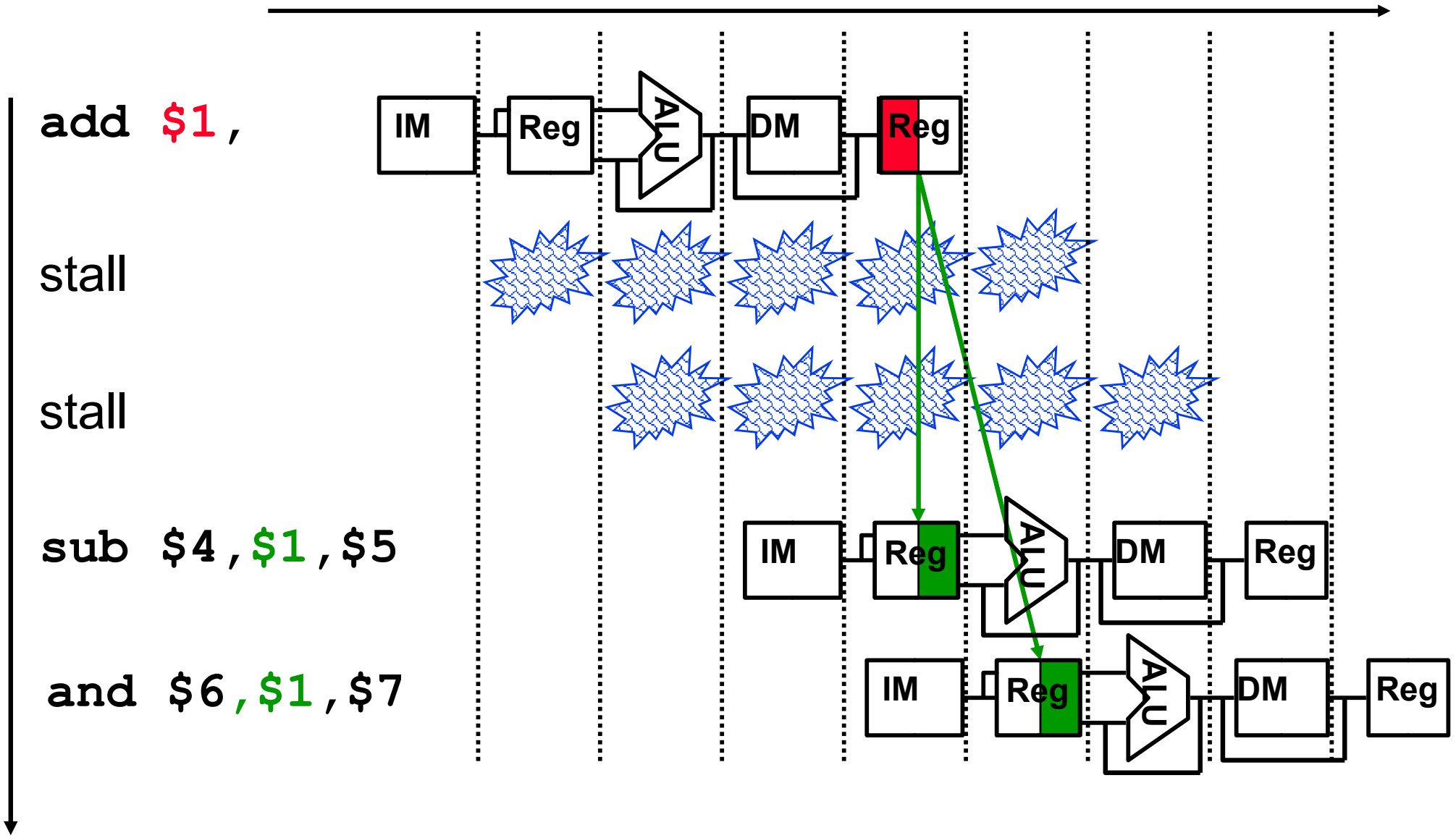
```
lw   $1,4($2)
sub  $4,$1,$5
and  $6,$1,$7
or   $8,$1,$9
xor  $4,$1,$5
```



Come evitare Data Hazard

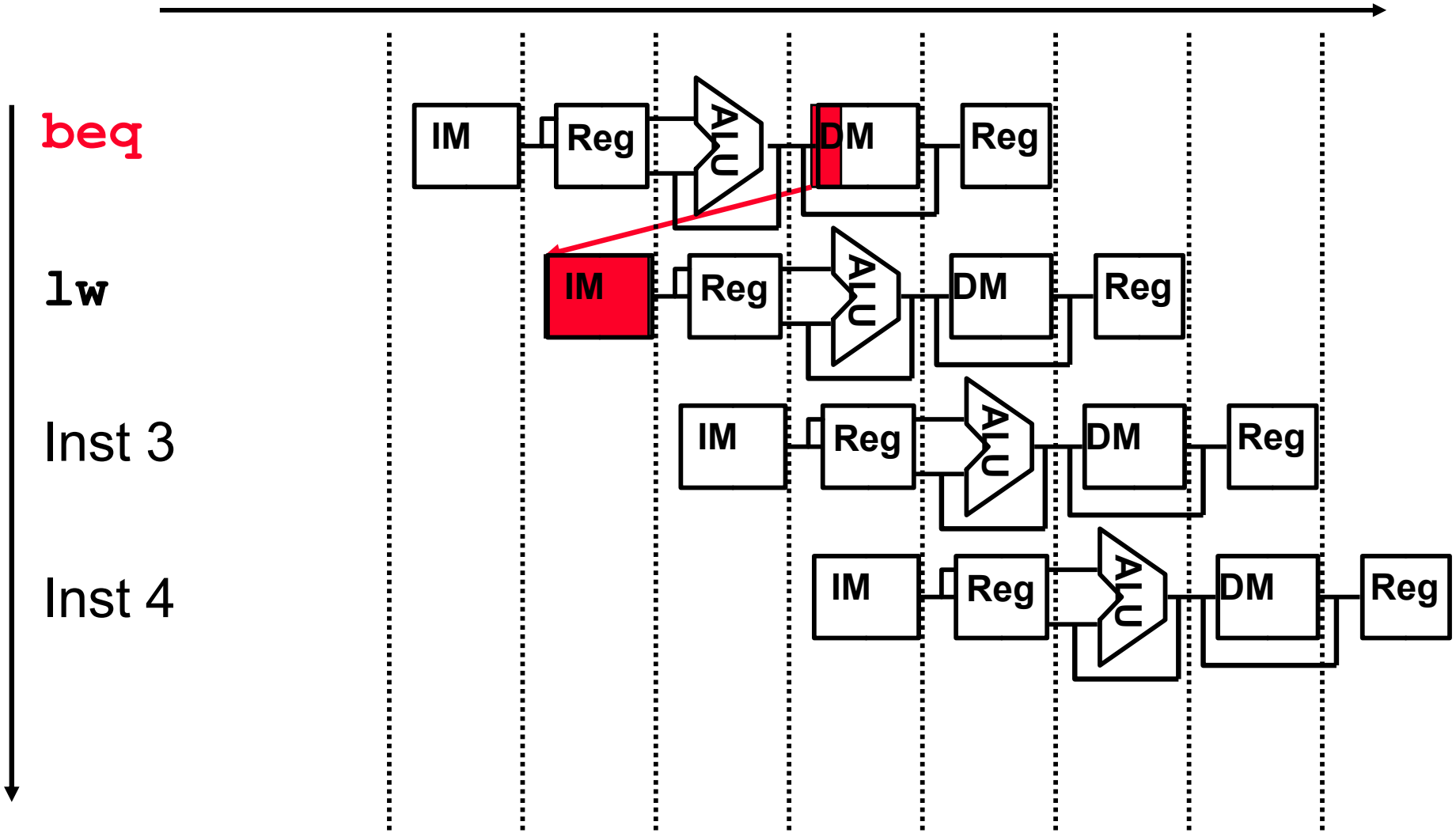


Come evitare Data Hazard

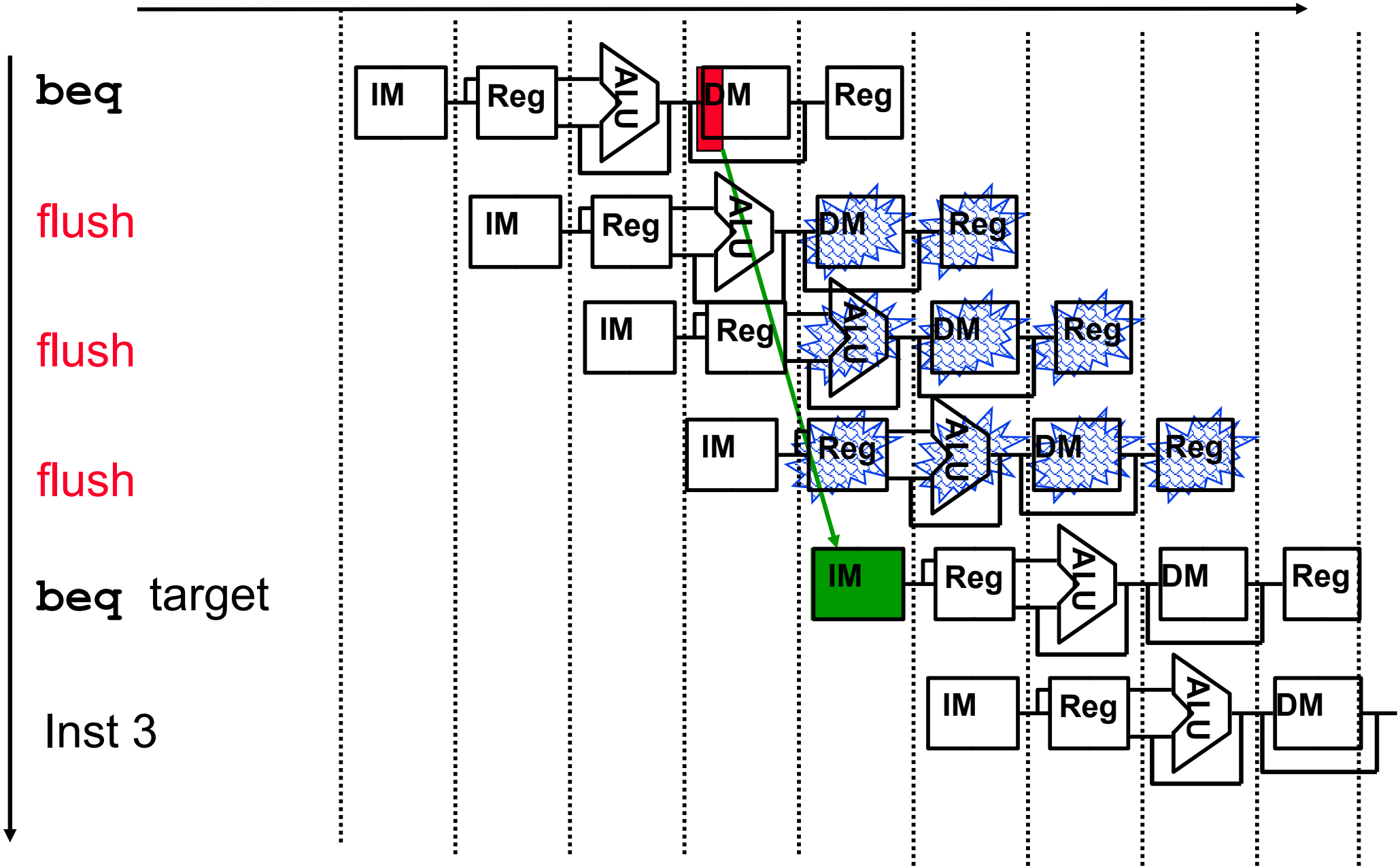


Control Hazards

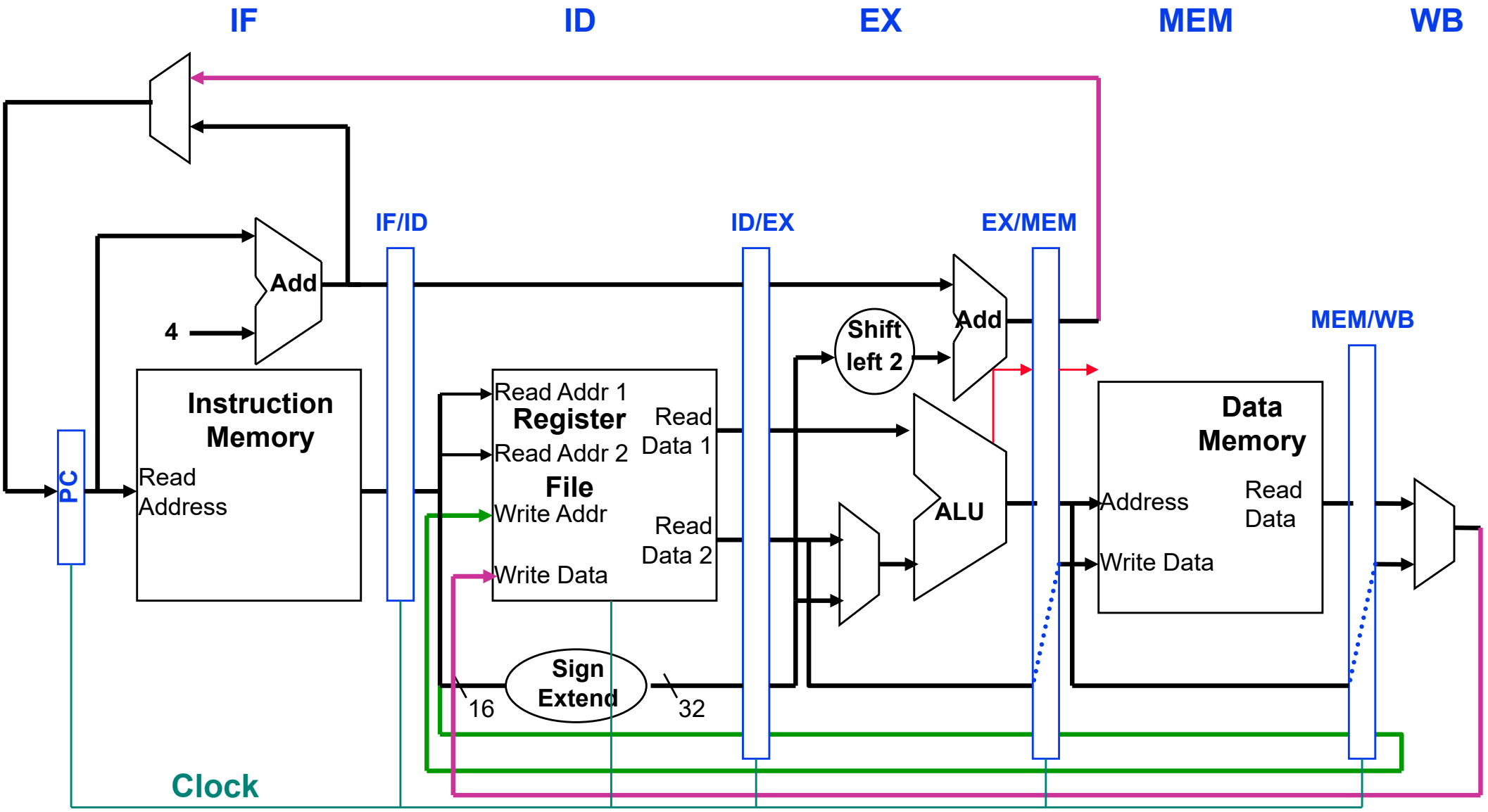
Le dipendenze causano conflitti (scritttura/lettura PC)



Come evitare control hazard



Pipeline Datapath



Altre possibili strutture per la pipeline

